INCH-POUND

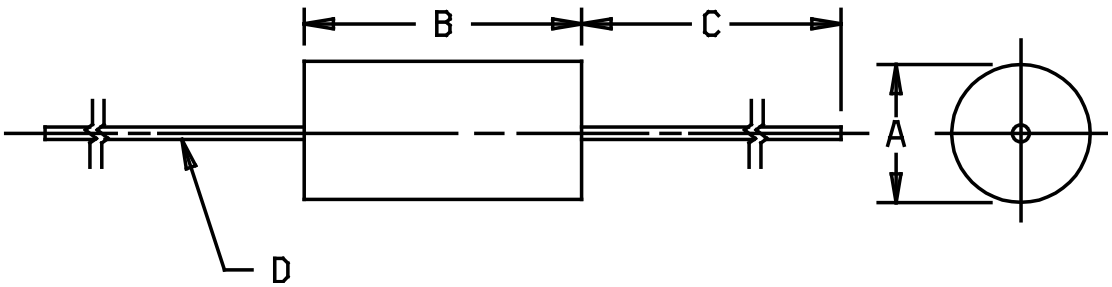
MIL-PRF-39010/2E
W/AMENDMENT 2
14 December 2022
SUPERSEDING
MIL-PRF-39010/2E
W/AMENDMENT 1
6 September 2018

PERFORMANCE SPECIFICATION SHEET

COILS, FIXED, RADIO FREQUENCY, MOLDED, MICROMINIATURE, MAGNETICALLY SHIELDED (IRON CORE-IRON SLEEVE), ESTABLISHED RELIABILITY AND NONESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39010.



Letter	Dimensions are in inches with metric equivalents (mm) in parentheses	Dimensions are in inches with metric equivalents (mm) in parentheses
	Minimum	Maximum
A	.152 (3.86)	.172 (4.37)
B	.390 (9.91)	.430 (10.92)
C	1.250 (31.75)	1.626 (41.30)
D	.023 (0.58)	.027 (0.69)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 1. Dimensions and configuration.

AMSC N/A

FSC 5950

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REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Material: Iron core with iron sleeve.

Weight: 0.04 ounce maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: MIL-STD-202-301; test voltage of 1,000 V rms.

Barometric pressure: MIL-STD-202-105; test condition C, 70,000 feet with a test voltage of 200 V rms.

Electrical characteristics: See table I and table II.

Inductance: See table I.

Inductance tolerance: See table I.

Q values: See table I.

Self-resonant frequency: See table I.

Percent coupling: 3 percent maximum.

DC resistance: See table I.

Temperature rise: 15°C.

Terminal pull: 5 pounds.

Part or Identifying Number (PIN): M39010/02 (dash number from table I).

Supersession data: This specification supersedes MS75088.

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TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q minimum	Test frequency (MHz)	Self-resonant frequency minimum (MHz)	DC resistance (25°C) maximum (ohms)	Rated dc current (mA) 2/
AR91**	.91	5	44	25	140	.07	1,070
A1R0**	1.00	5, 10	44	25	140	.07	1,070
A1R1**	1.10	5	44	7.9	130	.10	895
A1R2**	1.20	5, 10	44	7.9	130	.10	895
A1R3**	1.30	5	44	7.9	115	.12	815
A1R5**	1.50	5, 10	44	7.9	115	.12	815
A1R6**	1.60	5	44	7.9	105	.14	755
A1R8**	1.80	5, 10	44	7.9	105	.14	755
A2R0**	2.00	5	44	7.9	100	.19	650
A2R2**	2.20	5, 10	44	7.9	100	.19	650
A2R4**	2.40	5	44	7.9	92	.28	535
A2R7**	2.70	5, 10	44	7.9	92	.28	535
A3R0**	3.00	5	44	7.9	85	.35	480
A3R3**	3.30	5, 10	44	7.9	85	.35	480
A3R6**	3.60	5	44	7.9	75	.40	450
A3R9**	3.90	5, 10	44	7.9	75	.40	450
A4R3**	4.30	5	44	7.9	70	.55	380
A4R7**	4.70	5, 10	44	7.9	70	.55	380
A5R1**	5.10	5	44	7.9	65	.72	335
A5R6**	5.60	5, 10	44	7.9	65	.72	335
A6R2**	6.20	5	44	7.9	55	1.02	280
A6R8**	6.80	5, 10	50	7.9	55	1.02	280
A7R5**	7.50	5	50	7.9	50	1.32	245
A8R2**	8.20	5, 10	50	7.9	50	1.32	245
A9R1**	9.10	5	50	7.9	46	1.62	220
A100**	10.00	5, 10	50	7.9	46	1.62	220
A110**	11.0	5	50	2.5	44	2.00	200
A120**	12.0	5, 10	50	2.5	44	2.00	200

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the product level (e.g., C, M, P, R, S) and will be added to the end of the dash number.

2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

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TABLE II. Electrical characteristics (final). 1/

Inspection group	Allowable variation from initial measurement	Allowable variation from initial measurement	Allowable percent from specified minimum value in electrical characteristics (initial) table	Allowable percent from specified minimum value in electrical characteristics (initial) table
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±5			-10
Group IV	±5	±(3% +.001 ohm)	2/	-10
Group V	±5	±(2% +.001 ohm)	2/	-10
Quality conformance inspection Group B				
Subgroup 1	±5	±(2% +.001 ohm)	2/	-10
Subgroup 3	±5			-10
Subgroup 4	±5	±(3% +.001 ohm)	2/	-10

1/ Test fixture allowance of +.01 μH shall be added to all change in inductance limits ±(_ percent +.01 μH).
2/ The self-resonant frequency shall be not less than the value specified in table I.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-39010, this document references the following:

MIL-STD-202-105, MIL-STD-202-301

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

Review activities:
Army - AR, CR4, MI
Navy - AS, CG, MC, OS, SH
Air Force - 19

(Project 5950-2022-070)

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